DO NOW — RECALL

Final mixed-paper readiness

Name: _____ Class: _____ Date: _____

1. Difference between MAR and MDR?

2. How to encode a negative integer in two's complement?

3. What does O(n^2) describe?

4. Full adder A=1 B=1 carry-in=1. State sum and carry.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

MAR holds address; MDR transports data/instruction contents.

2. Model answer

Invert fixed-width positive representation then add one.

3. Model answer

Quadratic asymptotic growth.

4. Model answer

1+1+1=3=11_2; sum1 carry1.

By the end, you can…

• Apply Paper 1 and Paper 2 knowledge together

• Demonstrate precise independent working

• Identify the strongest remaining revision priority

Vocabulary — know these terms

readiness — Ability to apply without prompts.

synoptic — Knowledge across multiple topics.

method evidence — Reasoning supporting an answer.

NEW LEARNING

Read, annotate and explain

Identify

Retrieve knowledge across both components

Model

Show working in numeric and algorithm tasks

Apply

Check terminology and boundary cases

Evaluate

Prioritise a final independent improvement target

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

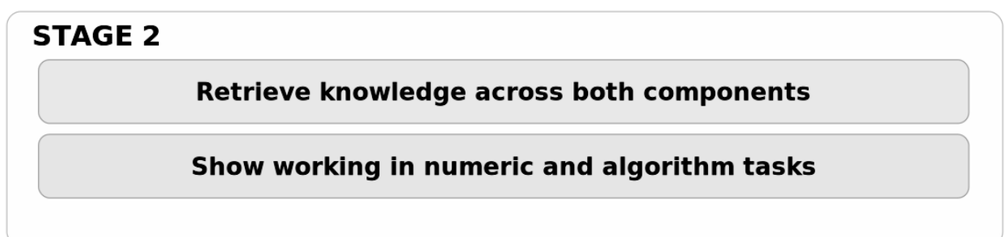
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



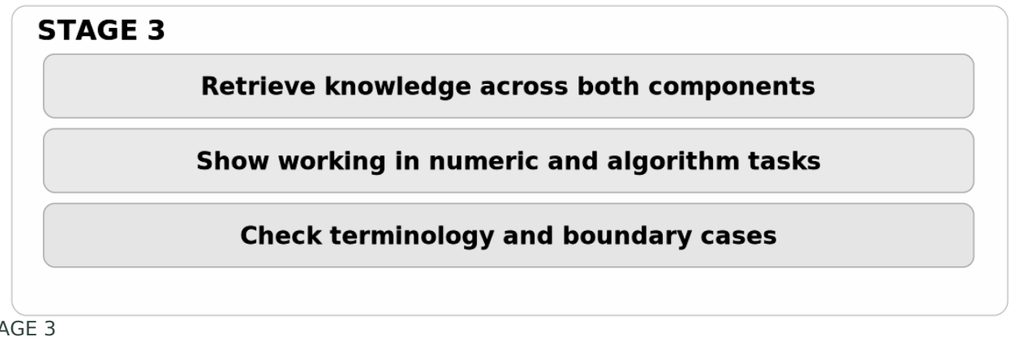
STAGE 1

New Learning



STAGE 2

New Learning



STAGE 3

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Search 9 in [1,3,5,7,9] using binary search.

Each midpoint comparison reduces the interval; return index4.



WEDO · MODEL

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

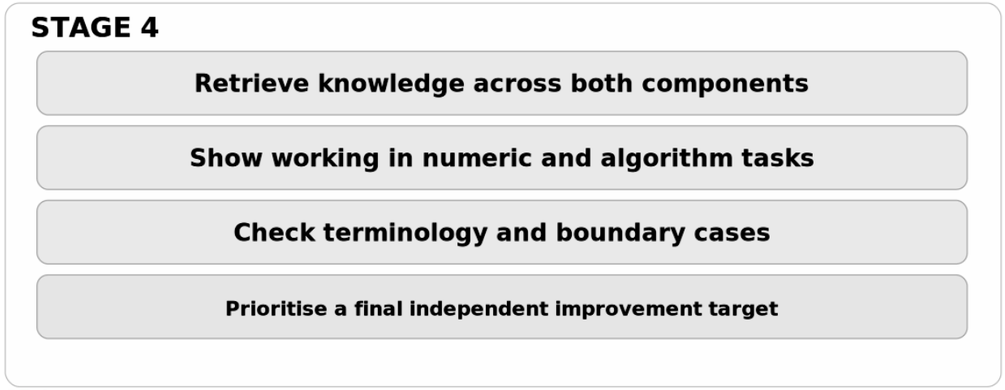
Guided 1 — CHECK

Explain the method
Each midpoint comparison reduces the interval; return index4.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



STAGE 4

I DO — WORKED EXAMPLE

Follow the worked method

Normalise positive $0.0101_2 \times 2^3$.

Explain the method

Shift mantissa one position left and reduce exponent by one.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

Publishing every lesson guarantees each student is ready for assessment.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

Q3 [9]: Evaluate cloud storage versus USB-only storage for an inclusive school revision service; weigh access, reliability, security. [9]

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Normalise positive $0.0101_2 \times 2^3$.

Shift mantissa one position left and reduce exponent by one.



IDO · MODEL

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

Search 9 in [1,3,5,7,9] using binary search.
Source code / input:
mid2→5, low3; mid3→7, low4; mid4→9 found.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Q1 [8]: In a simplified sequential fetch, PC=12 and memory address 12 stores instruction ADD 7. State MAR, MDR, CIR and next PC (assume no branch or interrupt). [4] Convert −5 to 8-bit two's complement. [4]

Question 2

Q2 [8]: Trace binary search for 11 in [1,3,5,7,9,11,13]. [4] Explain base case and recursion progress. [4]